

PRITIKA PRIYADARSHINI, San Francisco, CA
Linkedin: <https://www.linkedin.com/in/pritika-priyadarshini>
Portfolio: <https://pritika.studio/>

Email : pritikaapriyadarshini@gmail.com
Phone: +1 9452492481
Github: <https://github.com/pritika292>

SUMMARY

Full-stack/backend engineer with 5 years of experience building scalable distributed systems, real-time platforms, and developer tooling. Strong zero-to-one development experience, backend architecture, and owning ambiguous problems through production. I work best in high-ownership teams that value speed, reliability, and end-to-end execution.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, C++, PHP

Backend: Node.js, Express.js, FastAPI, REST, GraphQL, gRPC, Protobuf, Kafka, WebSockets, SSE

Frontend: React, Redux Toolkit, Next.js, Vite, Tailwind CSS, HTML, CSS

AI / LLM: LLM agents, MCP, function calling, RAG, pgvector, prompt shields, Copilot CLI, Claude Code

Data: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch

Cloud / Infra: AWS, Azure, Kubernetes, Docker, GitHub Actions, CI/CD, IAM, Lambda, S3, SQS, SNS

Testing / Tools: PyTest, Vitest, Playwright, pre-commit, gitleaks, Splunk, Git, Jira, Confluence

WORK EXPERIENCE

Paycom

San Francisco, CA

Software Engineer

June 2024 - Present

- Built Paycom's real-time messaging platform from zero to 100K+ users in a Kubernetes-based microservices environment, owning system design, database schema, REST APIs, and React/TypeScript dashboards end to end.
- Reduced database load by about 40% and improved peak-time responsiveness by integrating Redis Cache and a Kafka-based event-driven service to power real-time messaging dashboards and conversation updates.
- Built a reliable SMS integration layer with idempotent sends, retry-safe delivery, and provider callback handling to prevent duplicate messages and keep delivery status accurate at scale.
- Built a scalable lookup service to handle nested employee-group condition sets, replacing expensive multi-table joins with batch-updated mappings and reducing repeated high-volume query costs by ~60% across core modules.
- Designed and implemented long-running jobs for bulk updates and scheduled communications using asynchronous queue-based processing, improving system reliability by off-loading heavy operations out of user-facing request flows.
- Built responsive user interfaces for desktop and mobile using ReactJS, TypeScript, and Redux RTK, implementing reusable components and efficient state management while seamlessly integrating with backend APIs.
- Built an autonomous CI debugging agent with GitLab MCP and LLM tool-calling to classify CI failures, identify root causes from logs and Splunk, detect cross-PR infra issues, and auto-fix common pipeline failures.

Paycom

Irving, TX

Software Engineer Intern

May 2023 – August 2023

- Developed a real-time chat feature which connects more than 37,000 clients with their specialists. Designed MySQL database for secure storage and optimized querying of multiple message threads and tracking history.
- Implemented REST APIs using Node.js, Express, and Typescript. Leveraged Socket.io for instant messaging capabilities and delivered a user-friendly front-end using ReactJS.

VMware

Bangalore, India

Software Engineer

July 2020 – July 2022

- Built Java/REST API automation for VMware NSX-T software-defined networking, validating platform behavior across multi-release test matrices and catching ~50% of NSX regressions before production release.
- Owned pipeline-as-code CI/CD automation for vSphere regression testing, triggering test matrices on every build, enforcing promotion gates for release branches, and integrating Jira dashboards for pipeline health, test results, and on-call alerting.
- Reduced full-regression runtime (10h to 2h) by sharding Python test matrices across an autoscaling worker pool with queue-based scheduling, container-isolated execution, and shared setup caching.
- Enabled parallel regression testing by replacing per-worker database setup with a shared backing database, isolated by per-run namespaces and transaction-level rollback, reducing setup overhead while preserving test isolation.

VMware

Software Engineer Intern

Bangalore, India

January 2020 – July 2020

- Developed an internal tool using Python and Shell scripts to dynamically generate and execute test suites for custom upgrade/patch workloads. Track status and notify users with results on run completion, reducing hours of manual work.

Microsoft

Software Engineer Intern

Hyderabad, India

May 2019 – July 2019

- Built a pipeline flow to persist test results from applications running on Docker and displayed them on Azure DevOps platform using ReactJS.
- Implemented a clustering algorithm in C# to group related test failures and publish to Azure DevOps platform.

PROJECTS (<https://pratika.studio/>)

Shared Infra: Azure VM, Docker, docker-compose, Caddy, Azure Key Vault, GitHub Actions, OIDC

Focus Room | Live: <https://focusroom.pritika.studio/> | Source: <https://github.com/pratika292/focusroom>

Stack: TypeScript, Express, React+ Vite, Postgres, Azure OpenAI (gpt-4.1-mini), Azure Prompt Shields, SSE, Vitest

- Built a real-time AI audience simulator that streams threaded reactions from 20 diverse personas for user-submitted taglines, hot takes, and questions.
- Implemented concurrent multi-agent orchestration, SSE pub/sub streaming, layered prompt-injection defense, and keyless Azure authentication with DefaultAzureCredential.

Postgres-Inspector | Live: <https://pg.pritika.studio/> | Source: <https://github.com/pratika292/pg-inspector>

Tech: TypeScript, Express 5, React + Vite, Postgres, Redis, react-flow, pgsql-ast-parser, Azure OpenAI (gpt-4.1-mini)

- Built a PostgreSQL sandbox with 5 realistic schemas, visualizing ER diagrams, generating SQL from natural language, analyzing EXPLAIN plans, and surfacing AI-assisted query optimization suggestions to improve latency.

Control Room | Live: <https://controlroom.pritika.studio/> | Source: <https://github.com/pratika292/controlroom>

Tech: Python, FastAPI, Express, React, Postgres, Redis, Docker, Caddy

- Built an ops dashboard for the pratika.studio project family with real-time service status, deploy history, health-check streams, uptime tracking, and incident history across public projects.
- Implemented background health checks, SSE-based live updates, PostgreSQL-backed SLO views, shared Caddy routing with Let's Encrypt, and Azure service, token usage, VM, and cost visibility.

ShortLive | Live: <https://shortlive.pritika.studio/> | Source: <https://github.com/pratika292/shortlive>

Stack: TypeScript, Express, React, Postgres, Redis, WebSocket, Recharts, Vitest

- Built a real-time URL shortener with sub-second analytics by keeping redirects on a fast indexed lookup + 302 path while asynchronously queuing click events for background processing.
- Implemented Redis-backed event streaming with PostgreSQL persistence, WebSocket fan-out to active dashboards, and reliable HMAC-signed webhooks with retries and DLQ handling.

EDUCATION

University of Texas at Dallas

Master of Science in Computer Science; GPA 3.88/4.0

Richardson, TX

August 2022 – May 2024

Manipal Institute of Technology

Bachelor of Technology in Computer Science; GPA 9.02/10.0

Manipal, India

August 2016 – July 2020